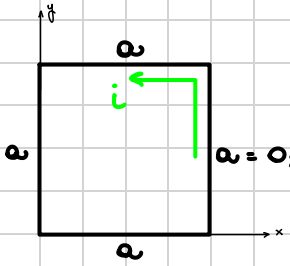
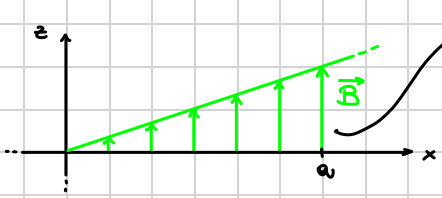
# Esercizi scheda 7

## Esercizio 1




$i = 5A$ $\qquad \vec{B} = \left(0{,}2\,\frac{T}{m}\right) x\,\hat{z}$




Il campo magnetico va da 0 per $x=0$ al valore $\vec{B}(a) = \alpha a\,\hat{z} = (0{,}04\,T)\,\hat{z}$

Sul lato della spira posto lungo l'asse y il campo magnetico è nullo. ⇒ La forza esercitata su quel lato della spira è pari a 0.

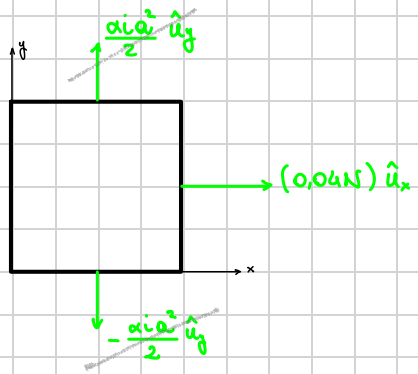
Sul lato della spira parallelo all'asse y si ha: $\vec{F} = i\vec{\ell} \times \vec{B} = (ia)\,\hat{y} \times B\,\hat{z} = (Bai)\,\hat{x} = (0{,}04\,N)\,\hat{x}$

Sul lato posto sull'asse x. $\quad d\vec{F} = i\,d\vec{\ell} \times d\vec{B}(x) = (i\,dx)\,\hat{x} \times (B(x))\,\hat{z} = -(\alpha x \cdot i\,dx)\,\hat{y}$

$$\Rightarrow \vec{F} = -\left(\int_0^a \alpha i x\,dx\right)\hat{y} = -\alpha i\left(\int_0^a x\,dx\right)\hat{y} = -\alpha i\left[\frac{x^2}{2}\right]_0^a \hat{y} = -\alpha i\,\frac{a^2}{2}\,\hat{y}$$

Analogamente, sul lato parallelo all'asse x si ottiene $\vec{F} = \alpha i\,\frac{a^2}{2}\,\hat{y}$

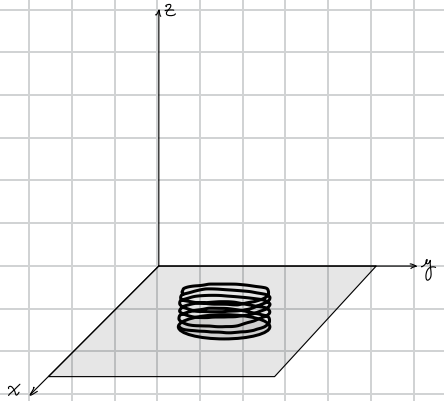
Complessivamente si ha quindi:




$\vec{F}_{TOT.} = (0{,}04\,N)\,\hat{x}$

## Esercizio 2

$N = 100$ spire $\quad R = 10\,cm = 0{,}1\,m \quad i = 8A \qquad \vec{B} = (0{,}6T)\,\hat{x} + (-0{,}4T)\,\hat{y} + (0{,}2T)\,\hat{z}$




Per una spira. $\vec{m} = iS\,\hat{n}$ ⇒ Per una bobina: $\vec{m} = NiS\,\hat{n}$

La bobina giace sul piano $xy \Rightarrow \hat{n} = \hat{z} \quad \rightarrow \vec{m} = Ni\pi R^2\,\hat{z} = (25{,}133\,Am^2)\,\hat{z}$

$$\vec{M} = \vec{m} \times \vec{B} = [(Ni\pi R^2)\,\hat{z}] \times [(0{,}6T)\,\hat{x} + (-0{,}4T)\,\hat{y} + (0{,}2T)\,\hat{z}]$$

$$= [(0{,}6T)(Ni\pi R^2)]\,(\hat{z} \times \hat{x}) - [(0{,}4T)(Ni\pi R^2)]\,(\hat{z} \times \hat{y}) + [(0{,}2T)(Ni\pi R^2)]\,\cancel{(\hat{z} \times \hat{z})}^{\,0}$$

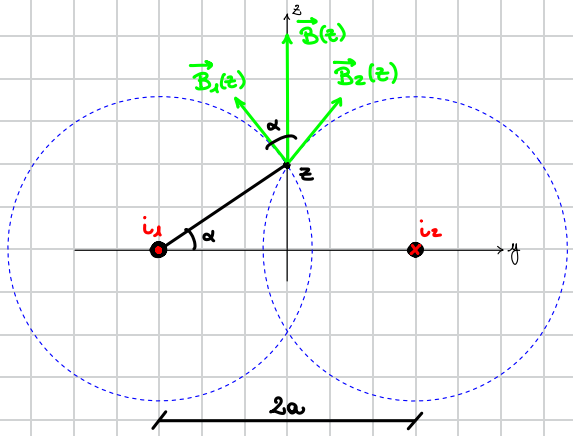
$$= [(0{,}6T)(Ni\pi R^2)]\,\hat{y} - [(0{,}4T)(Ni\pi R^2)]\,(-\hat{x})$$

$$= (10{,}053\ Nm)\,\hat{y} + (15{,}08\ Nm)\,\hat{x}$$

$$U = -\vec{m}\cdot\vec{B} = -(Ni\pi R^2)\,\hat{z}\cdot[(0{,}6T)\hat{x} + (-0{,}4T)\hat{y} + (0{,}2T)\hat{z}] \qquad \hat{x}\cdot\hat{z} = \hat{y}\cdot\hat{z} = 0, \quad \hat{z}\cdot\hat{z} = 1$$

$$= -(0{,}2T\ Ni\pi R^2) = -5{,}027\,J$$

## Esercizio 3




$2a = 4\,cm \Rightarrow a = 2\,cm = 0{,}02\,m$

$i_1 = i_2 = i = 50\,A$

Non ho capito nemmeno io a cosa servono sti dati...

Visto il problema simmetrico, $\vec{B}(z)$ è posto lungo l'asse z (le componenti lungo y si annullano) ed è pari a due volte la componente lungo z di $\vec{B}_1(z)$ o $\vec{B}_2(z)$.

$$B(z) = 2(\vec{B}_1(z)\cdot\hat{z}) = 2B_1(z)\cos\alpha = \cancel{2}\cdot\mu_0\frac{i}{\cancel{2}\pi\sqrt{z^2+a^2}}\cdot\frac{a}{\sqrt{z^2+a^2}} = \mu_0\frac{ia}{\pi(z^2+a^2)}$$

$$\Rightarrow \vec{B}(z) = \mu_0\frac{ia}{\pi(a^2+z^2)}$$